

# WEEK 2 POWER BI PROJECT REPORT

## DATA MODELING, DAX & SALES DASHBOARD

### 1. Introduction

In Week 2, the cleaned Superstore dataset prepared during Week 1 was used to develop a structured data model and sales dashboard using Microsoft Power BI.

The main focus of this week was to create Customer and Product tables, establish relationships, create DAX measures, and develop visualizations for analyzing sales, profit, orders and quantity.

### 2. Objectives

The main objectives are:

- To create Customer and Product tables.
- To remove duplicate records from the dimension tables.
- To establish relationships between the tables.
- To create a dedicated \_Measures table.
- To create DAX measures for business analysis.
- To create KPI cards.
- To analyze sales by category.
- To analyze sales trends by year.
- To analyze profit by category.
- To analyze profit by region.
- To develop a Power BI dashboard.

### 3. Creating Customer and Product Tables

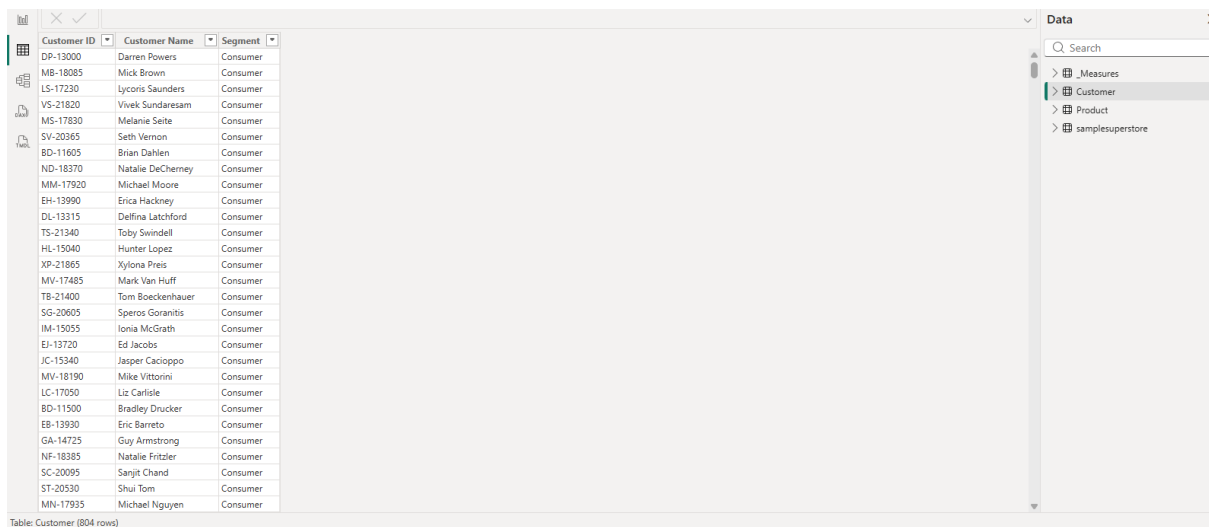
To organize the Superstore data, two additional tables named **Customer** and **Product** were created from the samplesuperstore table.

## Customer Table

The Customer table contains:

- Customer ID
- Customer Name
- Segment

Duplicate Customer IDs were removed so that each customer appears only once in the Customer table.



| Customer ID | Customer Name     | Segment  |
|-------------|-------------------|----------|
| DP-13000    | Darren Powers     | Consumer |
| MB-18085    | Mick Brown        | Consumer |
| LS-17230    | Lycoris Saunders  | Consumer |
| VS-21820    | Vivek Sundaresham | Consumer |
| MS-17830    | Melanie Seite     | Consumer |
| SV-20365    | Seth Vernon       | Consumer |
| BD-11605    | Brian Dahlen      | Consumer |
| ND-18370    | Natalie DeCherney | Consumer |
| MM-17920    | Michael Moore     | Consumer |
| EH-13990    | Erica Hackney     | Consumer |
| DL-13315    | Delfina Latchford | Consumer |
| TS-21340    | Toby Swindell     | Consumer |
| HL-15040    | Hunter Lopez      | Consumer |
| XP-21865    | Xylona Preis      | Consumer |
| MV-17485    | Mark Van Huff     | Consumer |
| TB-21400    | Tom Boeckenhauer  | Consumer |
| SG-20605    | Speros Goranitis  | Consumer |
| IM-15055    | Ionia McGrath     | Consumer |
| EJ-13720    | Ed Jacobs         | Consumer |
| JC-15340    | Jasper Cacioppo   | Consumer |
| MV-18190    | Mike Vittorini    | Consumer |
| LC-17050    | Liz Carlisle      | Consumer |
| BD-11500    | Bradley Drucker   | Consumer |
| EB-13930    | Eric Barreto      | Consumer |
| GA-14725    | Guy Armstrong     | Consumer |
| NF-18385    | Natalie Fritzier  | Consumer |
| SC-20095    | Sangit Chand      | Consumer |
| ST-20530    | Shui Tom          | Consumer |
| MN-17935    | Michael Nguyen    | Consumer |

Table: Customer (804 rows)

## Product Table

The Product table contains:

- Product ID
- Product Name
- Category
- Sub-Category

Duplicate Product IDs were removed to create a unique Product table.

| Product ID      | Category        | Sub-Category | Product Name                                                                 |
|-----------------|-----------------|--------------|------------------------------------------------------------------------------|
| OFF-PA-10000174 | Office Supplies | Paper        | Message Book, Wirebound, Four 5 1/2" X 4" Forms/Pg., 200 Dupl. Sets/Book     |
| OFF-PA-10000955 | Office Supplies | Paper        | Southworth 25% Cotton Granite Paper & Envelopes                              |
| OFF-PA-10002005 | Office Supplies | Paper        | Xerox 225                                                                    |
| OFF-PA-10002947 | Office Supplies | Paper        | Xerox 1923                                                                   |
| OFF-PA-10000659 | Office Supplies | Paper        | TOPS Carbonless Receipt Book, Four 2-3/4 x 7-1/4 Money Receipts per Page     |
| OFF-PA-10002693 | Office Supplies | Paper        | Wirebound Service Call Books, 5 1/2" x 4"                                    |
| OFF-PA-10002689 | Office Supplies | Paper        | Weyerhaeuser First Choice Laser/Copy Paper (20Lb. and 88 Bright)             |
| OFF-PA-10001804 | Office Supplies | Paper        | Xerox 195                                                                    |
| OFF-PA-10000380 | Office Supplies | Paper        | REDIFORM Incoming/Outgoing Call Register, 11" X 8 1/2", 100 Messages         |
| OFF-PA-10003072 | Office Supplies | Paper        | Euroka Recycled Copy Paper 8 1/2" x 11", Ream                                |
| OFF-PA-10001204 | Office Supplies | Paper        | Xerox 1972                                                                   |
| OFF-PA-10000350 | Office Supplies | Paper        | Message Book, Standard Line "While You Were Out", 5 1/2" X 4", 200 Sets/Book |
| OFF-PA-10000466 | Office Supplies | Paper        | Memo Book, 100 Message Capacity, 5 3/8" x 11"                                |
| OFF-PA-10003363 | Office Supplies | Paper        | Xerox 204                                                                    |
| OFF-PA-10001800 | Office Supplies | Paper        | Xerox 220                                                                    |
| OFF-PA-10000069 | Office Supplies | Paper        | TOPS 4 x 6 Fluorescent Color Memo Sheets, 500 Sheets per Pack                |
| OFF-PA-10003036 | Office Supplies | Paper        | Black Print Carbonless 8 1/2" x 8 1/4" Rapid Memo Book                       |
| OFF-PA-10000327 | Office Supplies | Paper        | Xerox 1971                                                                   |
| OFF-PA-10000587 | Office Supplies | Paper        | Array Parchment Paper, Assorted Colors                                       |
| OFF-PA-10001878 | Office Supplies | Paper        | Xerox 1891                                                                   |
| OFF-PA-10002259 | Office Supplies | Paper        | Geographics Note Cards, Blank, White, 8 1/2" x 11"                           |
| OFF-PA-10000213 | Office Supplies | Paper        | Xerox 198                                                                    |
| OFF-PA-10002137 | Office Supplies | Paper        | Southworth 100% Résumé Paper, 24lb.                                          |
| OFF-PA-10000176 | Office Supplies | Paper        | Xerox 1887                                                                   |
| OFF-PA-10001215 | Office Supplies | Paper        | Xerox 1963                                                                   |
| OFF-PA-10002120 | Office Supplies | Paper        | Xerox 1889                                                                   |
| OFF-PA-10000919 | Office Supplies | Paper        | Xerox 1918                                                                   |
| OFF-PA-10002195 | Office Supplies | Paper        | Xerox 1966                                                                   |
| OFF-PA-10002365 | Office Supplies | Paper        | Xerox 1967                                                                   |

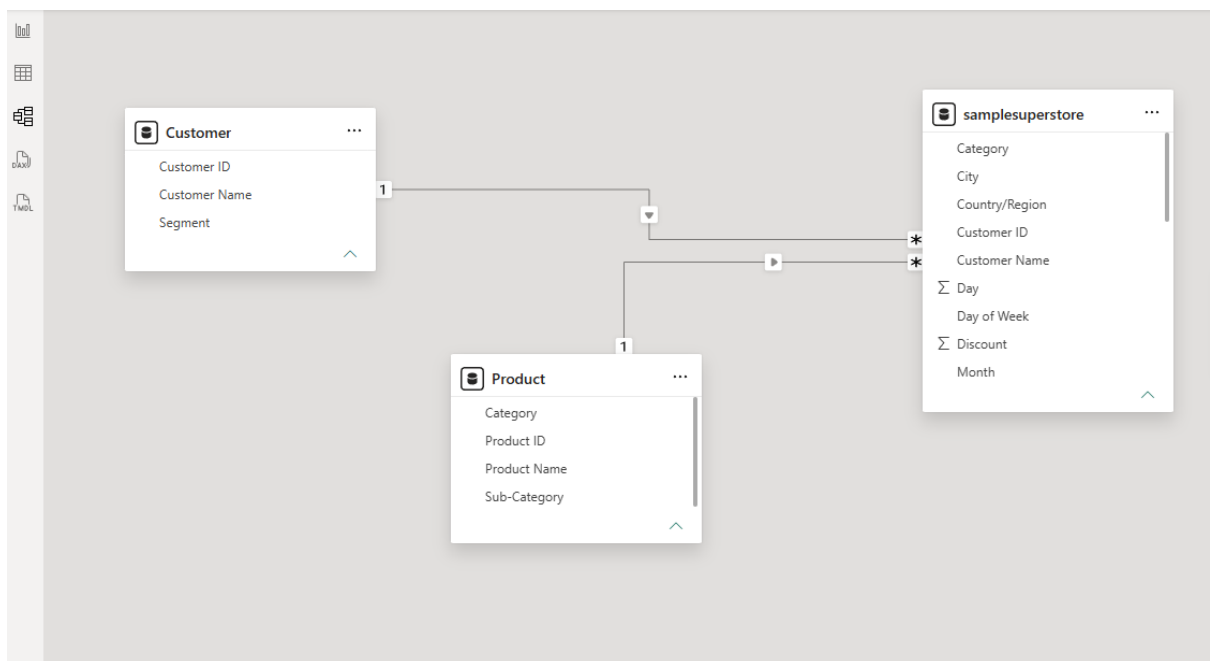
Table: Product (1,862 rows)

## 4. Data Modeling and Relationships

After creating the Customer and Product tables, relationships were established with the main **samplesuperstore** table.

The **Customer** and **Product** tables act as dimension tables, while **samplesuperstore** contains the main transactional data.

These relationships help organize the data and allow the dashboard visuals to work correctly with the related tables.



## 5. Creating the \_Measures Table

A dedicated table named \_Measures was created to store the DAX measures.

This table was created to keep all the measures organized in one location.

The following measures were created:

- Total Sales
- Total Profit
- Total Quantity
- Total Orders
- Average Sales
- Profit Margin %

## 6. DAX Measures

DAX measures were created to calculate important business performance indicators.

### Total Sales

Total Sales = SUM('samplesuperstore'[Sales])

This measure calculates the total sales generated from all transactions.

### Total Profit

Total Profit = SUM('samplesuperstore'[Profit])

This measure calculates the total profit generated from the transactions.

### Total Quantity

Total Quantity = SUM('samplesuperstore'[Quantity])

This measure calculates the total quantity of products sold.

### Total Orders

Total Orders = DISTINCTCOUNT('samplesuperstore'[Order ID])

This measure calculates the number of unique orders in the dataset.

### **Average Sales**

Average Sales = AVERAGE('samplesuperstore'[Sales])

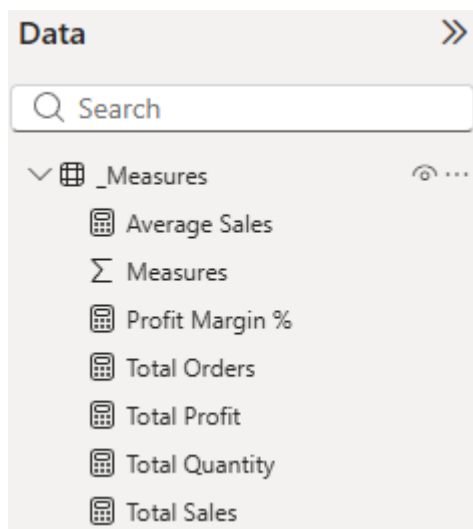
This measure calculates the average sales value across the transactions.

### **Profit Margin %**

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

This measure calculates the percentage of sales represented by profit.

## **DAX Measures Created in Power BI**



## **7. KPI Cards**

KPI cards were created to provide a quick overview of important business performance indicators.

The KPI cards include:

- Total Sales
- Total Profit

- Total Orders
- Total Quantity
- Average Sales
- Profit Margin %

The cards provide a simple summary of the overall business performance.



## 8. Total Sales by Category

A column chart was created to compare total sales across different product categories.

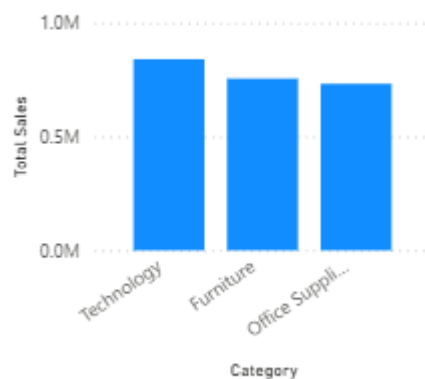
The categories analyzed were:

- Technology
- Furniture
- Office Supplies

The visualization helps compare the sales contribution of each category.

Based on the visualization, Technology has the highest sales among the three categories.

**Total Sales by Category**



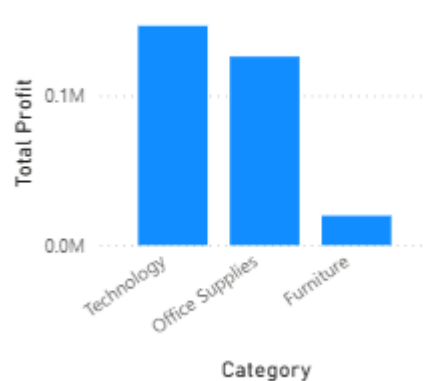
## 9. Total Profit by Category

A column chart was created to compare total profit across different product categories.

The visualization helps identify the contribution of each category to overall profitability.

Based on the visualization, Technology has the highest profit among the categories shown.

**Total Profit by Category**



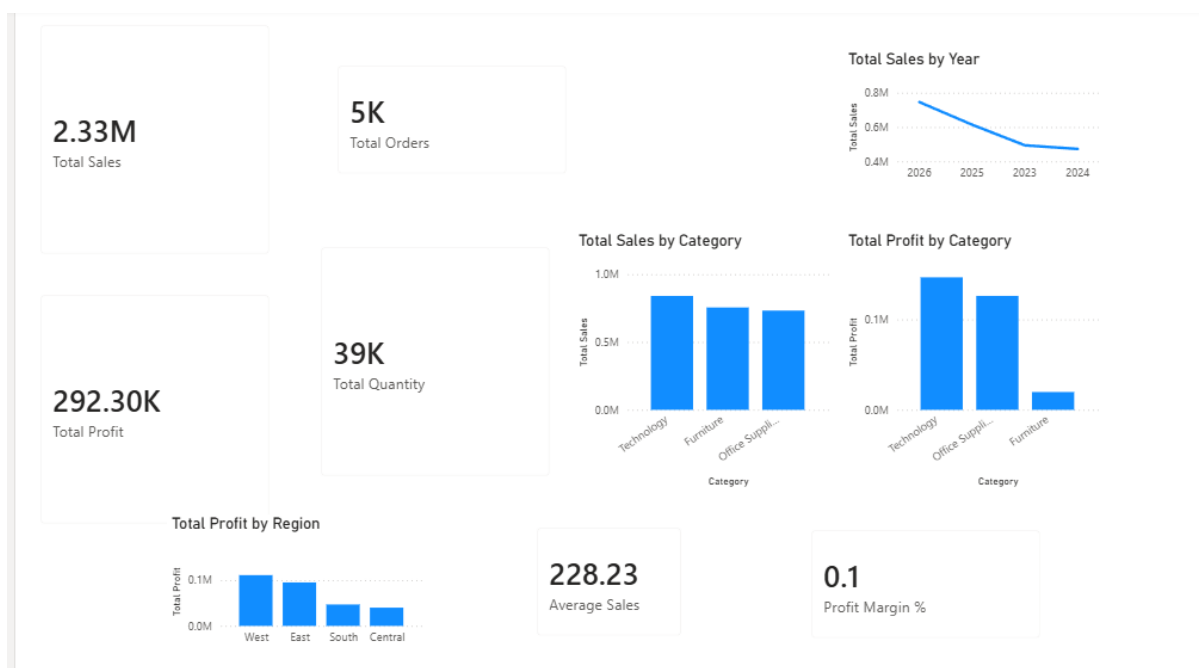
## 10. Final Dashboard

The KPI cards and visualizations were arranged together to create the final Superstore Sales Dashboard.

The dashboard provides an overview of:

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Average Sales
- Profit Margin %
- Sales by Category
- Profit by Category

The dashboard makes it easier to understand the overall sales and profit performance of the Superstore dataset.



## 11. Findings

Based on the dashboard analysis, the following findings were observed:

1. Technology has the highest sales among the three major categories.
2. Technology also contributes the highest profit among the categories shown.
3. The KPI cards provide a quick overview of overall business performance.

4. The DAX measures help calculate important business indicators such as sales, profit, orders and quantity.
5. The dashboard presents the business information in a simple and visual format.

## **12. Conclusion**

Week 2 successfully developed the Superstore Power BI project by creating a structured data model, DAX measures and a dashboard.

Customer and Product tables were created from the main Superstore table, and relationships were established between the tables. A dedicated \_Measures table was also created to organize the DAX calculations.

Six DAX measures were created to calculate Total Sales, Total Profit, Total Quantity, Total Orders, Average Sales and Profit Margin %. KPI cards and charts were then developed to analyze the sales and profit performance of different categories.

The final Power BI dashboard provides a clear and visual representation of the business performance of the Superstore dataset.